```
#include<stdio.h>
#define MAX 100
int main(){

int A[20][20],B[20][20];
int tupleA[MAX][3],tupleB[MAX][3],tupleC[MAX][3],tupleT[MAX][3];
int r,c;
int i,j,k;

printf("enter the number of rows and columns of matrix:");
scanf("%d%d",&r,&c);

printf("Enter the elements of A matrix:");
for(i=0;i<r;i++){
for(j=0;j<c;j++){
scanf("%d",&A[i][j]);
}
}

printf("Enter the elements of B matrix:");
for(i=0;i<r;i++){
for(j=0;j<c;j++){
scanf("%d",&B[i][j]);
}
}

//TUPLE OF MATRIX A
k=1;
tupleA[0][0]=r;
tupleA[0][1]=c;

for(i=0;i<r;i++){
for(j=0;j<c;j++){
        if(A[i][j]!=0){
        tupleA[k][0]=i;
        tupleA[k][1]=j;
        tupleA[k][2]=A[i][j];
        k++;

        }
}
}
tupleA[0][2]=k-1;
printf("tuple representation of A:\n");
printf("row\tcolumn\tvalue\n");
for(i=0;i<=tupleA[0][2];i++){
printf("%d\t%d\t%d\n",tupleA[i][0],tupleA[i][1],tupleA[i][2]);
}


//TUPLE OF MATRIX B
k=1;
tupleB[0][0]=r;
tupleB[0][1]=c;

for(i=0;i<r;i++){
for(j=0;j<c;j++){
        if(B[i][j]!=0){
        tupleB[k][0]=i;
        tupleB[k][1]=j;
        tupleB[k][2]=B[i][j];
        k++;

        }
        }
        }
```

```
tupleB[0][2]=k-1;

printf("tuple representation of B:\n");
printf("row\tcolumn\tvalue\n");
for(i=0;i<=tupleB[0][2];i++){
printf("%d\t%d\t%d\n",tupleB[i][0],tupleB[i][1],tupleB[i][2]);
}


//ADDITION OF TUPLE A AND B

i=1;
j=1;
k=1;

tupleC[0][0]=r;
tupleC[0][1]=c;

while(i<=tupleA[0][2] && j<=tupleB[0][2]){
if(tupleA[i][0]==tupleB[j][0] && tupleA[i][1]==tupleB[j][1] ){
     tupleC[k][0]=tupleA[i][0];
     tupleC[k][1]=tupleA[i][1];
     tupleC[k][2]=tupleA[i][2]+tupleB[j][2];
     i++;
     j++;
      k++;
}
else if(tupleA[i][0]< tupleB[j][0] || tupleA[i][0]==tupleB[j][0] && tupleA[i][1]<tupleB[j]
[1]){
        tupleC[k][0]=tupleA[i][0];
     tupleC[k][1]=tupleA[i][1];
     tupleC[k][2]=tupleA[i][2];
     i++;
     k++;

}
else{
        tupleC[k][0]=tupleB[j][0];
     tupleC[k][1]=tupleB[j][1];
     tupleC[k][2]=tupleB[j][2];
     j++;
     k++;

}
}

while(i<=tupleA[0][2]){
tupleC[k][0]=tupleA[i][0];
     tupleC[k][1]=tupleA[i][1];
     tupleC[k][2]=tupleA[i][2];
     i++;
     k++;
     }

while(j<=tupleB[0][2]){
        tupleC[k][0]=tupleB[j][0];
     tupleC[k][1]=tupleB[j][1];
     tupleC[k][2]=tupleB[j][2];
     j++;
     k++;
     }


tupleC[0][2]=k-1;
```

```
 printf("tuple Addition of A and B:\n");
printf("row\tcolumn\tvalue\n");
for(i=0;i<=tupleC[0][2];i++){
printf("%d\t%d\t%d\n",tupleC[i][0],tupleC[i][1],tupleC[i][2]);
}


//Transpose

tupleT[0][0]=tupleC[0][1];
    tupleT[0][1]=tupleC[0][0];
    tupleT[0][2]=tupleC[0][2];

    k=1;

        for(int col=0;col<c;col++)
        {
        for(i=1;i<=tupleC[0][2];i++)
        {
        if(tupleC[i][1]==col)
        {
        tupleT[k][0]=tupleC[i][1];
        tupleT[k][1]=tupleC[i][0];
        tupleT[k][2]=tupleC[i][2];
        k++;
        }
        }
        }

    // Display Transpose Tuple
    printf("\nTranspose of Resultant Tuple\n");
    printf("Row\tCol\tValue\n");
    for(i=0;i<=tupleT[0][2];i++)
        printf("%d\t%d\t%d\n",tupleT[i][0],tupleT[i][1],tupleT[i][2]);


return 0;
}
```